

CAMERA-EQUALIZED ADVERSARIAL TRAINING FOR CROSS-VIEW URBAN OBJECT RE-IDENTIFICATION

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ABSTRACT

The Urban Elements Re-Identification Challenge 2026 evaluates queries from a single backward-facing camera (c004) while the training set contains only forward-facing cameras (c001–c003), with no cross-view identity pairs. External data provides a strong initial lift, but on top of it Domain-Adversarial Neural Network (DANN) training is limited by the same asymmetry: under random batch sampling, nearly half of batches contain no c004 image and the expected c004 count is only ≈ 0.86 out of 32 images, so the Gradient Reversal Layer receives a thin cross-view signal. We address this with a Camera-Equalized Sampler that guarantees a c004 image whenever a cross-view identity is drawn. A paired ablation isolates $+0.008$ mAP from the sampler independently of DANN, and $+0.005$ mAP from DANN on top. With camera-aware re-ranking, class filtering, checkpoint averaging and query flipping, the system reaches Kaggle public mAP 0.175, currently 5th on 49% of the test set.

Index Terms— Urban re-identification, domain-adversarial training, batch sampling, cross-view retrieval, Part-Aware Transformer

1. INTRODUCTION

Urban element re-identification (ReID) matches infrastructure objects across camera views and supports infrastructure monitoring in smart cities [1]. The Urban Elements ReID Challenge 2026 [2] introduces a structural viewpoint asymmetry: the 11,175 training images come from three forward-facing cameras (c001–c003), but the 928 evaluation queries come from a single backward-facing camera (c004) that traverses the route in reverse, and the competition training set contains no forward–backward identity pairs.

Building on the Part-Aware Transformer (PAT) [3] baseline, we observe that Domain-Adversarial Neural Network (DANN) [4] training does close part of this gap, but its effect is bounded by how often the minority camera appears in a training batch. Under standard random sampling, c004 is missing from nearly half of batches, and the Gradient Reversal Layer (GRL) is reduced to generic regularisation of the forward views. We give a probabilistic account of this behaviour, introduce a Camera-Equalized Sampler that structurally restores c004 exposure, and provide a controlled paired ablation that isolates the contribution of the sampler from that of DANN.

2. PROPOSED METHOD

2.1. Training data and augmentations

We merge the competition training set (11,175 images, 1,088 identities, c001–c003 only) with all three splits of the external dataset [1] (8,695 images, 691 identities, c001–c004), giving **19,870 images / 1,779 identities**. The external data is the only source of forward–backward pairs: 1,262 c004 images from 322 cross-view identities. Identity labels are kept disjoint via fixed offset.

We disable the Local Grayscale Transformation augmentation of the PAT baseline, which discards identity-discriminative colour

(sign background, container markings). Colour jitter (± 0.15 brightness/contrast, ± 0.1 saturation and hue, prob. 0.5) and Random Patch Token replacement (prob. 0.5) are retained across all experiments to absorb illumination drift and partial occlusions.

2.2. Backbone and training schedule

We use PAT [3] with a Vision Transformer (ViT) Large backbone (24 blocks, 1024-dim, 306M parameters, input 256×128). Training: Stochastic Gradient Descent (SGD), lr 0.008, weight decay 10^{-4} , cosine schedule, 5-epoch warm-up, 60 epochs, batch size 32 (8×4). The retrieval descriptor is the classification ([CLS]) token before the bottleneck batch-normalisation layer; the patch embedding from the released PAT checkpoint is kept frozen for training stability.

2.3. Camera-Equalized DANN

Adversarial head. A three-layer MLP ($1024 \rightarrow 512 \rightarrow 256 \rightarrow 4$, batch norm, no dropout) is attached to the [CLS] token through a Gradient Reversal Layer [4]. The total loss is $\mathcal{L} = \mathcal{L}_{\text{ReID}} + \mathcal{L}_{\text{cam}}$; the GRL scales $\mathcal{L}_{\text{cam}}$ gradients by $-\lambda_{\text{eff}}$ during backpropagation only. We use the standard schedule [4], $\lambda_{\text{eff}} = \lambda_{\text{max}}(2/(1 + e^{-10p}) - 1)$, $p = \text{step}/\text{total steps}$, $\lambda_{\text{max}} = 0.3$.

Why random sampling underperforms. With 322 c004-capable identities out of 1,779, drawing 8 identities uniformly leaves $\approx 20\%$ probability of no c004-capable identity in the batch; within a c004-capable identity only ≈ 4 of ≈ 27 images are c004, giving a further $\approx 50\%$ conditional miss. Marginalising over both, $\approx 47\%$ of batches contain no c004 image and the expected c004 count is only ≈ 0.86 out of 32 batch images. The GRL still receives forward-camera gradient, explaining DANN’s modest gain in Table 1, but the camera classifier accuracy (cam-acc) rises monotonically to 0.77 (Fig. 1): suppression along the c001–c003-vs-c004 axis is weak.

Camera-Equalized Sampler. For every c004-capable identity drawn in a batch, one of its four sampled images is forced to be a c004 image; the other three are sampled from its forward views. Forward-only identities are sampled unchanged. This raises the fraction of batches with ≥ 1 c004 image from $\approx 53\%$ to $\approx 79\%$ and the expected c004 count from ≈ 0.86 to ≈ 1.45 per batch, without altering the identity distribution ($\approx 61\%$ of batch identities still come from the competition set).

2.4. Post-processing pipeline

Class-aware gallery filtering and bin–container merge. The gallery is restricted to same-class images. Containers and rubbish bins are merged into a single `bin_container` super-class: they are visually indistinguishable at survey distances and separating them introduces false negatives.

Camera-Aware Jaccard re-ranking. CA-Jaccard [5] separates intra- and inter-camera k -reciprocal neighbourhoods, well suited where same-camera matches are by construction wrong. We use $k_{\text{intra}}=k_{\text{inter}}=10$, $k_2=4$; $k_1=10$ outperforms $k_1=15$ by $+0.005$ mAP.

Table 1. Training ablation. mAP uses standard k-reciprocal re-ranking ($k_1=20$, $k_2=6$, $\lambda=0.3$) as in the official PAT baseline; no class filtering, CA-Jaccard, ensembling or TTA. Rows 1–3 cover data/architecture scaling. Rows 4–7 form a paired study: rows 5 vs. 7 isolate the sampler under equal data and $\lambda_{\max}$; rows 6 vs. 7 isolate DANN under equal data and sampler.

Configuration	Data	Sampler / $\lambda_{\max}$	mAP
PAT, ViT-Base [3]	competition	random / –	0.102
+ external data (ViT-Base)	17.6 k	random / –	0.121
+ ViT-Large + –LGT	17.6 k	random / –	0.130
DANN, random sampling	17.6 k	random / 0.2	0.143
DANN, random sampling, all UAM splits	19.9 k	random / 0.3	0.144
Equalized sampling, no DANN	19.9 k	equalized / 0	0.152
Equalized sampling + DANN (final)	19.9 k	equalized / 0.3	0.157

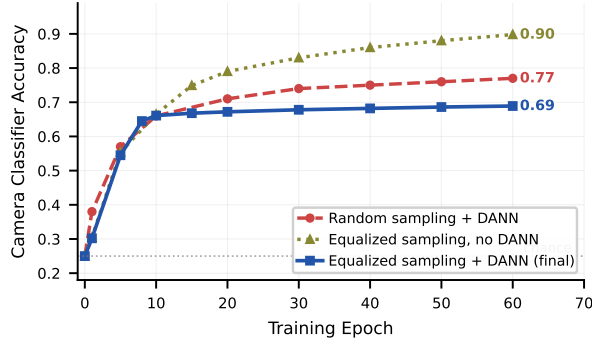


Fig. 1. Camera classifier accuracy (cam-acc) over training under three sampling/loss configurations. Random sampling with DANN (red) plateaus at 0.77 as the GRL receives only forward-camera gradient; equalized sampling without DANN (yellow) reaches 0.90 with the classifier unopposed; the final configuration (blue) settles at 0.69 with the GRL receiving and partially suppressing the backward-camera signal. Chance is 0.25.

Checkpoint ensembling. Image descriptors are computed as the unweighted L2-normalised average of the features extracted at epochs 50, 55, and 60.

Query-side test-time augmentation. Because c004 records the route in reverse, query objects appear horizontally mirrored relative to the gallery. We average each query descriptor with its horizontally-flipped version; gallery descriptors are not flipped.

3. EXPERIMENTS AND RESULTS

All experiments run on a single NVIDIA RTX 2080 Ti (11 GB). We report Kaggle public-leaderboard mAP@100, as local validation becomes uninformative once external splits are merged (training/validation identity overlap).

3.1. Training ablation

Table 1 reports the cumulative effect of each training change. The top block (rows 1–3) shows that data integration and backbone capacity are large levers — adding the external dataset contributes +0.019 at ViT-Base scale, and replacing ViT-Base with ViT-Large under LGT-off adds a further +0.009. The bottom block (rows 4–7) is the contribution of this paper. Rows 5 vs. 7 isolate the sampler: holding data and $\lambda_{\max}$ fixed and switching only random $\rightarrow$ equalized adds +0.013 mAP, while the same shift in data and $\lambda_{\max}$ under random sampling (rows 4 vs. 5) adds only +0.001. Rows 6 vs. 7 isolate DANN: the equalized sampler alone gives 0.152, an improvement of +0.008 over random sampling with DANN (row 5), and adding DANN on top of it gives a further +0.005. The two effects stack additively.

Table 2. Post-processing pipeline on the best training configuration of Table 1. Row 1 is the same number as the last row of Table 1, the starting point of the pipeline. Row 2 bundles three concurrent changes (re-ranker, class filter, super-class merge); rows 3–5 each add one component.

Post-processing step	mAP
k-reciprocal re-ranking (baseline, $k_1=20$, $k_2=6$)	0.157
$\rightarrow$ CA-Jaccard ($k_1=15$) + class filter + bin merge	0.166
+ tighter neighbourhood (CA-Jaccard $k_1=10$)	0.171
+ checkpoint ensemble (ep. 50/55/60)	0.173
+ query-side horizontal flip	0.175

The three cam-acc trajectories in Fig. 1 mirror this decomposition. The gap to chance (0.25) is expected: the ReID loss still encourages identity-discriminative features and backward views expose object surfaces not visible from the front, so camera and identity remain partially entangled.

3.2. Post-processing: which steps pay off

Two design choices drive most of the post-processing gain (Table 2). First, switching to a camera-aware re-ranker matters because every query is c004 and every gallery image is not, so generic k-reciprocal neighbourhoods are dominated by within-camera matches that are wrong by construction; replacing it with CA-Jaccard [5] together with same-class restriction and a bin-container merge adds +0.009. Second, tightening k_1 from 15 to 10 adds +0.005: with a sparse gallery (≈ 2.6 images per identity per camera), looser neighbourhoods admit too many false matches. Checkpoint averaging and query flipping add +0.002 each.

Approaches that did not help. View-specific embeddings, deterministic c004 oversampling ($n=2, 4$), and stochastic perspective augmentation each reduced or tied mAP: the first cannot generalise across the full route reversal, the second over-represents the external dataset domain, and the third destroys the geometric cues the camera classifier needs.

4. CONCLUSION

DANN-based camera alignment for cross-view urban ReID requires explicit sampling control: when the query camera is sparsely present in batches, the GRL regularises only the well-represented cameras. The Camera-Equalized Sampler restores the c004 signal, adding +0.008 mAP independently of DANN and a further +0.005 mAP from adversarial training on top. With re-ranking, checkpoint averaging and query flipping, the system reaches mAP 0.175 (+71.7% over the PAT baseline), currently 5th on the public leaderboard (49% of test). Backward-camera scarcity remains the main limitation; synthetic back-view generation is a natural next step.

5. REFERENCES

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